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Experimental and thermodynamic assessment of the Gd–Ti system

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ABSTRACT

The phase equilibria of the binary Gd–Ti system were studied by combining electrostatic levitation of the melt with high energy synchrotron in situ X-ray diffraction at elevated temperatures. The coexistence of the phases including the liquid was directly proven in the temperature range between $T=1000$ and 1920 K. The Gd–Ti system is of monotectic type characterized by a miscibility gap in the liquid state above the monotectic temperature $T=1841\pm5$ K. Together with experimental data from thermo-analysis and microstructure investigation of as-cast samples an improved Gd–Ti phase diagram is presented and a thermodynamic description is derived.

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1. Introduction

Intermetallic compounds formed between rare-earth and transition metals are of particular interest regarding their magnetic properties like the $GdCo_{11}Ti$ phase in the ternary system Co–Gd–Ti system [1]. The knowledge of thermodynamic and thermophysical properties is of high practical importance in alloy design and processing. In this connection, accurate data on the binary sub-systems are indispensable.

Experimental assessments of the Gd–Ti system were published by Croeni et al. [2] and by Savitskii et al. [3]. Both reports concluded an eutectic type phase diagram without intermetallic phases and low solubilities in the elemental phases. Based on these experimental data a thermodynamic assessment was done by Murray [4]. As a result, the presence of a miscibility gap in the liquid was predicted, which is also cited by Massalski [5]. Recently, Schmitz et al. [6] reported a modified Gd–Ti phase diagram with a wider miscibility gap in the liquid based on their microstructure investigations of quenched samples from electromagnetic levitation experiments. All the reports exhibit conflicts in the data and suffer from the lack of reliable thermodynamic data for the phase equilibrium with the liquid phase.

The aim of this paper is to reinvestigate the binary Gd–Ti system, as suggested by Okamoto [7]. Reliable experimental data on the liquid state were obtained by combining container-less electrostatic levitation at elevated temperatures with in situ synchrotron X-ray diffraction. Based on the experimental data, an essentially improved Gd–Ti phase diagram will be presented together with a consistent thermodynamic description.

2. Experimental

For in situ X-ray diffraction investigations, alloy ingots of 200 mg mass with nominal compositions $Gd_{30}Ti_{70}$, $Gd_{40}Ti_{60}$, $Gd_{60}Ti_{40}$ were prepared by arc-melting proper quantities of pure Gd and Ti (Alfa Aesar 99.99 wt% purity) under an argon atmosphere with a Ti getter. The ingots were inverted and remelted several times to obtain evenly distributed elements in the sample. The sizes of the spherical shaped ingots were about 4 mm in diameter, which is optimal size for the applied levitation technique. The difference in sample mass after arc-melting was less than 0.1%. In situ high energy synchrotron X-ray diffraction (XRD) was

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performed in transmission geometry with a wave length $\lambda = 0.01249$ nm at the DORIS beam-line BW5 HASYLAB Hamburg. The samples were processed under high vacuum conditions in an electrostatic levitator that has been specially designed for performing diffraction experiments of melts [8]. Two lasers have been applied for heating the samples. The temperature was measured by a single-color pyrometer. The pyrometer values were corrected by adapting the melting temperature of the Gd-phase as determined from the in situ X-ray diffraction patterns to that of the differential thermal analysis. The accessible temperature range was between 1000 K and 1920 K. The scattered intensity was continuously recorded with acquisition time of 0.5 s by a two-dimensional X-ray detector (PerkinElmer) either at a constant temperature or during heating or cooling. The scattered intensity $I(q)$ as a function of the wave vector q ($q = 4\pi \sin\theta/\lambda$) was obtained by azimuthal integration using Fit2D software [9].

For the sessile drop investigations, alloy ingots of 800 mg mass with nominal compositions $Gd_{40}Ti_{60}$, $Gd_{60}Ti_{40}$, $Gd_{70}Ti_{30}$ were prepared by arc-melting proper quantities of pure Gd and Ti (Alfa Aesar 99.99 wt% purity) under an argon atmosphere with a Ti getter. The sessile drop investigations were carried out using the experimental setup described in details elsewhere [10]. The samples were heated up to 2073 K with heating rate of 6 K/min, and then solidified either by fast cooling rate of 200–300 K/min by switching off the heater or by a controlled slow cooling rate of 30 K/min. The melting and solidification behavior of the Gd–Ti samples on zirconia substrates was registered by a high-speed and high-resolution CCD camera. Cross-sections of the samples after the sessile drop experiments were cut for metallographic examination by polishing resin-mounted specimens with diamond pastes. The microstructure was analyzed by scanning electron microscopy (SEM) in the backscattered electron (BSE) mode using a Gemini-Zeiss microscope equipped with an energy-dispersive spectrometer (Bruker AXS) for composition analysis. The chemical composition of the phases was determined at 30 keV from the intensities of the Gd–L and Ti–K emission lines. Differential thermal analysis (DTA) up to 1950 K was carried out using a Setaram DTA92 (zirconia container) under helium atmosphere with a heating rate of 10 K/min.

3. Experimental results and discussion

3.1. In situ observation of phase equilibria and phase transitions at elevated temperatures

Fig. 1 shows DTA scans of different Gd–Ti alloys up to 1950 K. The DTA curves indicate the occurrence of four phase transformations upon heating. The transition temperatures are summarized

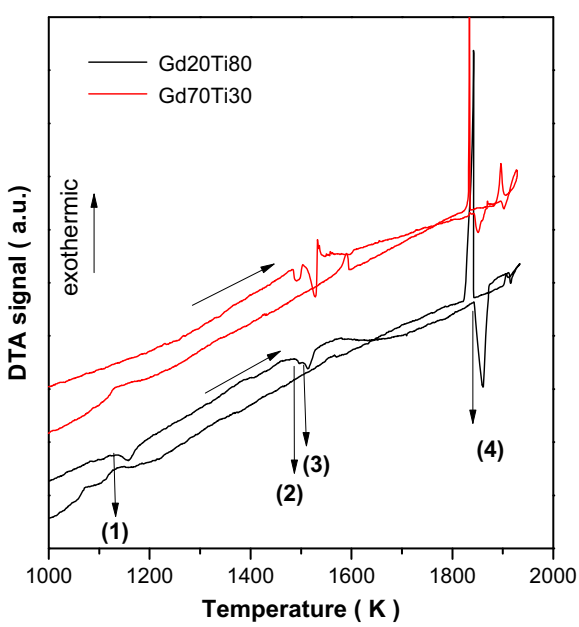

Fig. 1. DTA scans of $Gd_{20}Ti_{80}$ and $Gd_{70}Ti_{30}$ (10 K/min) up to 1900 K; heating and cooling curves are shown.

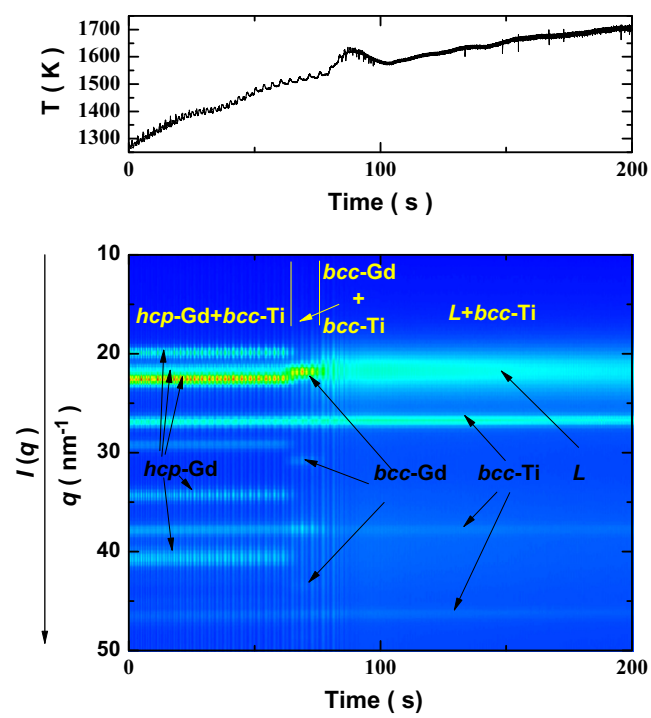

Fig. 2. Temperature profile (uncorrected pyrometer data) versus time and intensity map of XRD patterns $I(q)$ for $Gd_{30}Ti_{70}$. The patterns were recorded each for 0.5 s.

Table 1
Invariant reactions of the Gd–Ti system. Experimental data from DTA, in situ XRD and SEM are compared with the result of CALPHAD calculation using optimized parameters derived from the experiment.

Reaction	Experimental Gd (at%)			T(K)	Calculated Gd (at%)			T(K)
Monotectic e_1: $L_1 \leftrightarrow L_2$+*bcc*-Ti(Gd)	6.0 ± 1	76.0 ± 3	0.5 ± 0.2	1841 ± 5	7.5	75.0	1.0	1842
Eutectic e_2: L↔*bcc*-Gd(Ti)+*bcc*-Ti(Gd)	92 ± 3	99.5 ± 0.1	0.2 ± 0.1	1508 ± 5	95.4	0.8	99.4	1508
Eutectoid e_3: *bcc*-Gd(Ti)↔*hcp*-Gd(Ti)+*bcc*-Ti(Gd)	99.6 ± 0.1	99.9 ± 0.1	0.2 ± 0.1	1488 ± 5	99.4	0.8	99.8	1490
Eutectoid e_4: *bcc*-Ti(Gd)↔*hcp*-Gd(Ti)+*hcp*-Ti(Gd)	0.2 ± 0.1	99.9 ± 0.1	0.1 ± 0.1	1140 ± 5	0.3	99.9	0.1	1150

in Table 1. The DTA event (4) is indicated also in the cooling cycle; however the solidification of the Gd liquid as well as the *bcc*–*hcp* transformation of Gd are not clearly visible due to incomplete reaction of the melt with the zirconia crucible.

The corresponding phase transformations can be determined unambiguously from the in situ XRD patterns. Fig. 2 shows the temperature profile as a function of time during levitation of a $Gd_{30}Ti_{70}$ sample together with an intensity map of a series of simultaneously recorded diffraction patterns $I(q)$. The phases and the sequences of coexistence ranges within the temperature interval $T=1300$–1700 K are marked according to the characteristic diffraction peak positions. Fig. 3 shows selected XRD patterns $I(q)$ of a $Gd_{40}Ti_{60}$ sample recorded at different temperatures and representing the phase equilibria between $T=1460$ K and 1900 K. The whole set of recorded XRD patterns together with the onset temperatures of the DTA curves, enabled to determine the different coexistence ranges of phases as a function of temperature. Reactions (1) and (2) correspond to the phase transitions of the terminal solid phases. Below $T=1140$ K, *hcp*-Gd(Ti) and *hcp*-Ti(Gd) coexist. The reaction (1) of the DTA curve at $T=1140$ K corresponds to the eutectoid transformation *bcc*-Ti(Gd)↔*hcp*-Gd(Ti)+*hcp*-Ti(Gd). In the present study the determined temperature $T=1140$ K is lower than that of the *bcc*↔*hcp* transformation of pure Ti ($T=1155$ K [2]). This can be considered as indication to eutectoid character of this transformation. Between 1150 K $< T <$ 1480 K we observe the coexistence of the *hcp*-Gd(Ti) and *bcc*-Ti(Gd) phases. Reaction (2) at $T=1488$ K is then related to the eutectoid transformation *bcc*-Gd(Ti)↔*hcp*-Gd(Ti)+*bcc*-Ti(Gd). The coexistence of *bcc*-Gd(Ti) and *bcc*-Ti(Gd) is evident from the XRD patterns between 1490 K$\leq T \leq$1505 K.

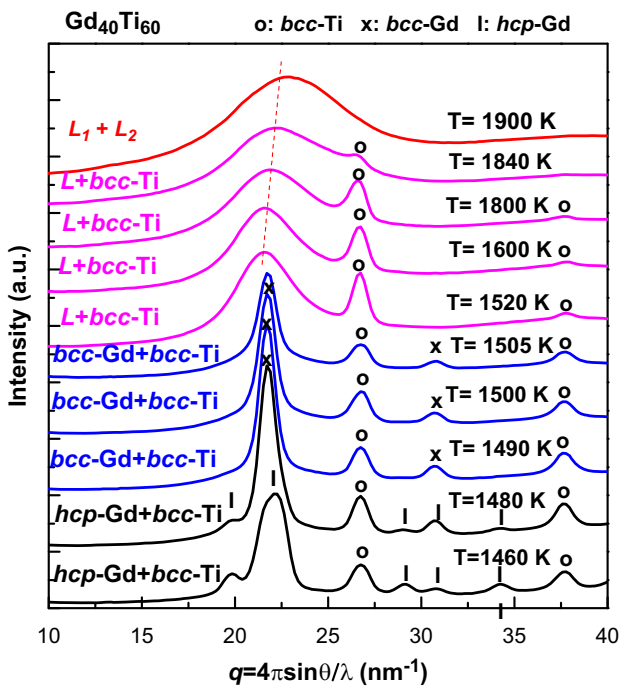

Fig. 3. XRD of $Gd_{40}Ti_{60}$ at elevated temperatures. Characteristic temperature range of phase coexistence and phase transitions is presented. Characteristic diffraction peaks of the solid phases are marked.

The DTA event (3) at $T=1508$ K reflects the melting of *bcc*-Gd(Ti) by the eutectic reaction L↔*bcc*-Gd(Ti)+*bcc*-Ti(Gd). The partial melting of the sample is indicated by the demising *bcc*-Gd(Ti) reflections and the occurrence of a diffuse diffraction maximum in the XRD-patterns. The in situ XRD patterns give clear evidence of coexistence of the Gd-rich melt L_2 and the *bcc*-Ti(Gd) phase in a wide temperature range from 1520 K up to about 1840 K. The same phase sequences as found for $Gd_{30}Ti_{70}$ (Fig. 2) and $Gd_{40}Ti_{60}$ (Fig. 3) were also observed in the in situ XRD patterns of the $Gd_{60}Ti_{40}$ alloy. Differences in the intensity ratios for the different alloy compositions are related to the volume fractions of the phases in accordance with the lever rule. Unfortunately, only a limited number of diffraction patterns could be obtained for $T>1840$ K. The reason is the large difference of mass density of the two Gd- and Ti-rich melts. Due to the decomposition reaction, the levitated samples became unstable in shape and position and then usually flew away. For the one sample $Gd_{40}Ti_{60}$, which could be measured at temperatures $T>1840$ K, we observe the disappearance of the *bcc*-Ti(Gd) phase reflections and simultaneously a larger shift of the position of the diffuse maximum (cf. Fig. 3). This behavior gives evidence for the occurrence of the monotectic reaction $L_1 \leftrightarrow L_2$+*bcc*-Ti(Gd) in accordance with the DTA event (4) at $T=1841$ K. The intensity contribution of the Ti-rich liquid L_1 is small and cannot be resolved even for the Ti-rich alloys, because Gd atoms scatter X-rays about 10 times stronger than Ti atoms. However, the shift of the maximum positions in Fig. 3 can only be explained by the change in composition of the Gd-rich liquid L_2 with temperature according to liquidus line and for $T\geq1840$ K along

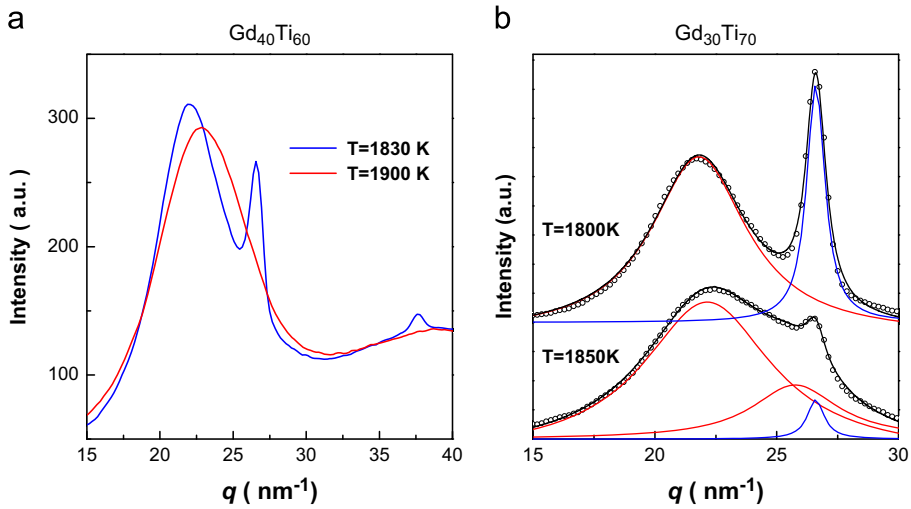

Fig. 4. Comparison of XRD patterns of Gd–Ti alloys below and above the monotectic temperature. (a) $Gd_{40}Ti_{60}$: a strong shift of the maximum position and a broader diffuse maximum are the indication for the presence of phase separation at $T=1900$ K. (b) $Gd_{30}Ti_{70}$: the change in the shape of the diffraction curve at $T=1850$ K is due to the decomposition into two liquids. The fit result of the experimental data (circles) by a sum of Lorentzian peak functions (lines) is shown. The XRD pattern of $Gd_{30}Ti_{70}$ at $T=1750$ K can only be fitted by a two phase liquid+small fraction of *bcc*-Ti. The contribution of the Ti-rich liquid is about three times stronger compared to that in $Gd_{40}Ti_{60}$.

the binodal line. The phase decomposition of the liquid phase leads also to a broadening of the diffuse diffraction maximum with some asymmetric part at the high-q side at $T \geq 1850$ K. Fig. 4(a) compares two diffraction patterns of $Gd_{40}Ti_{60}$ below and above the monotectic temperature. The shift of the maximum position and a broader diffuse maximum are the indication for the occurrence of phase separation at $T = 1900$ K. In Fig. 4(b) diffraction patterns are shown for a $Gd_{30}Ti_{70}$ alloy measured below and just above the monotectic reaction which was the highest temperature before the loss of the samples. The comparison clearly indicates the change of the shape due to the decomposition into two liquids much more clear because the intensity contribution of the Ti-rich liquid is about three times stronger in $Gd_{30}Ti_{70}$ compared to $Gd_{60}Ti_{40}$. The two characteristic experimental patterns (circles) were approximated by a sum of Lorentzian peak functions (line) as shown in Fig. 4(b). The XRD pattern of $Gd_{30}Ti_{70}$ at $T = 1750$ K cannot be fitted by only two curves as it is possible for that at $T = 1700$ K. There is still a small fraction of bcc-Ti present but the measured profile can only be described by two diffuse maxima as demonstrated in Fig. 4(b).

Fig. 5 shows in detail the position of the diffuse maximum q_1 of Gd-rich phase L_2 as a function of temperature. The position of the (110)-reflection of the bcc-Ti(Gd) phase $q_{(110)}$ is also given. The presented data are from independent measurements of the three Gd–Ti alloys having different compositions. The data exhibit a good reproducibility within some scatter band of $\Delta q_1 \pm 0.1$ nm^{-1}, which defines the accuracy of the experiment. The shift of $q_{(110)}$ to lower q-values in Fig. 5 reflects mainly the thermal lattice expansion of the *bcc*-Ti(Gd) phase. The maximum position q_1 of the Gd-rich liquid exhibits an opposite behavior, shifting to larger q-values, which is related to increasing solubility of Ti at higher temperatures. For metallic liquids and metallic glasses the position of the first diffuse maximum q_1 scales approximately with a concentration-averaged atomic diameter D of the alloy analogously to the Ehrenfest equation [11]:

$$q_1 = k2\pi/D \tag{1}$$

with a constant $k \approx 1.2$, and $D = \Sigma c_i D_i$, where D_i is the atomic diameter of element i and c_i is its concentration. This equation is not exactly valid, however, within an element substitution series

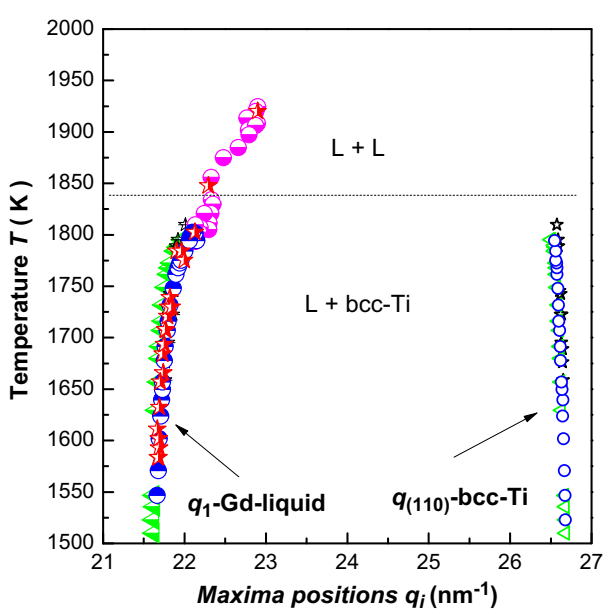

Fig. 5. Maximum positions q_1 of the Gd(Ti) liquid, and $q_{(110)}$ of *bcc*-Ti versus T data are from measurements of different Gd–Ti alloys (◁,◁: $Gd_{60}Ti_{40}$, ●,◒,○: $Gd_{40}Ti_{60}$, ★☆: $Gd_{30}Ti_{70}$).

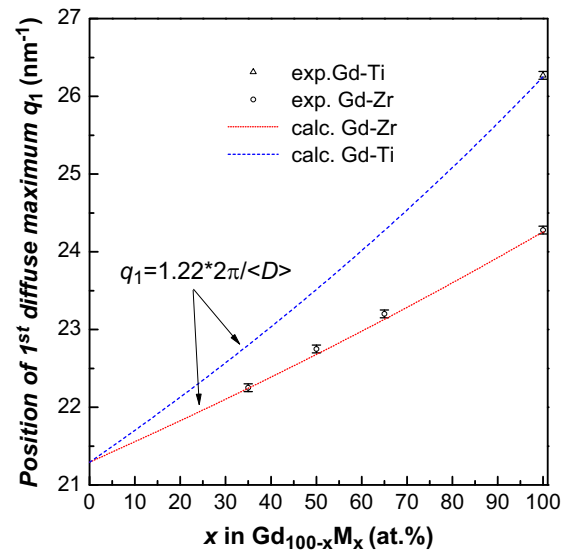

Fig. 6. Position of the 1st diffuse diffraction maximum of Gd–M liquids (M = Ti, Zr). Experimental data from Refs. [14–16] are given by symbols. Calculations based on the Ehrenfest equation [11] are shown by the lines.

the change in q_1 can be often well described by a linear dependence on the concentration c_i of the added element [12,13]. An increased solubility of Ti ($D_{Ti} = 0.292$ nm) in the Gd-rich liquid ($D_{Gd} = 0.360$ nm) reduces the average atomic diameter resulting in the experimentally observed shift of q_1 to larger q-values with rising temperature. The influence of the thermal expansion of the liquid on the position q_1 is one order of magnitude smaller but within the error bars of q_1 determination.

Fig. 6 compares experimental available data of the XRD diffraction maxima position of liquid Zr [14], Ti [15], and Gd–Zr [16] with the calculation by Eq. (1) with $k = 1.22$. There is a good agreement between calculation and experimental data for the Gd–Zr liquids. Also the value of q_1 for pure Ti is in accordance with the experimental one. Because Zr and Ti have quite similar chemical properties Eq. (1) should be also applicable to the Gd–Ti liquid. From a linear fit of the data at the Gd-rich side ($0 \leq x \leq 30$ at% Ti) we obtain

$$q_1(x) = 21.28 + 0.0424x. \tag{2}$$

Then, the maximum position q_1 of the liquid phase as shown in Fig. 5 can be used to determine the chemical composition of the Gd(Ti) liquid as a function temperature. The Ti-content x is calculated from the experimental data $q_1(T)$ with q_1 in nm^{-1} and x in at%:

$$x_{Ti}^{liquid}(T) = \frac{q_1(T) - 21.28}{0.0424} \tag{3}$$

Taking the q_1-values of the liquid diffraction maximum in the two phase region L+*bcc*-Ti(Gd), the calculations represent the liquidus line of the Gd–Ti phase diagram as shown in Fig. 7. The liquidus line ranges from about 92 ± 3 at% Gd for $T = 1500$ K up to about 80 ± 3 at% Gd for $T = 1800$ K. The presence of a rather steep liquidus line inferred from the position of maxima q_1 is in accordance with the weak change of the diffraction intensities between Gd-rich liquid phase L_2 and *bcc*-Ti(Gd) phase up to 1800 K as can be seen in Fig. 3. The error in determining the maxima position is about ± 0.1 nm^{-1}. From that an error of ± 2 at% follows for the calculated chemical composition of the Gd-rich liquid. For temperatures $T > 1841$ K, the shift of the maximum position and the derived melt composition correspond to the Gd-rich part of the binodal line.

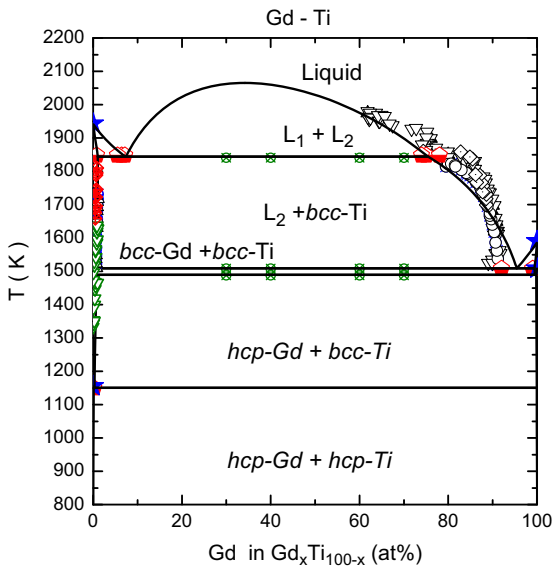

Fig. 7. Experimentally assessed Gd-Ti phase diagram. XRD-data : ○ $Gd_{30}Ti_{70}$, △ $Gd_{40}Ti_{60}$ (sample a), ▽ $Gd_{60}Ti_{40}$ (sample b); DTA scans : ⊠ ; SEM : ⬟; phase transformations of terminal phases:★ (taken from [5]). Lines represent the CALPHAD calculation of this work using the parameters given in Table 3.

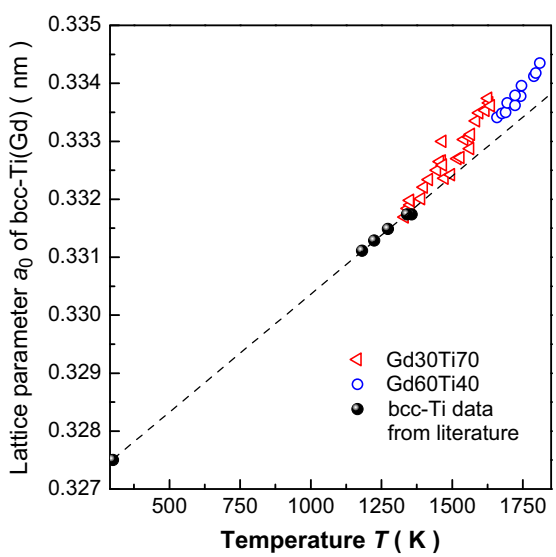

Fig. 8. Temperature dependence of the lattice parameter a_0 of *bcc*-Ti(Gd) (▽,○). The deviation from the values of pure *bcc*-Ti (● [17,18]) represented by the dashed line is due to dissolution of Gd.

The solubility of Gd in the *bcc*-Ti(Gd) phase can be estimated by the analysis of the temperature dependence of the lattice parameter $a_{bcc\text{-Ti(Gd)}}$ shown in Fig. 8. Using Vegard's rule for a solid solution phase the concentration of Gd (in at%) in the *bcc*-Ti(Gd) phase is given by

$$x_{Gd}(T) = \frac{a_{bcc-\mathrm{Ti(Gd)}}(T) - a_{bcc-\mathrm{Ti}}(T)}{a_{bcc-\mathrm{Gd}} - a_{bcc-\mathrm{Ti}}} 100. \quad (4)$$

The temperature dependence of the lattice parameter of the pure *bcc*-Ti phase, $a_{bcc\text{-Ti}}(T)$, can be approximated by a linear extrapolation $a = 0.32629(1+1.25\times10^{-5}T)$ nm using the literature data given in Refs. [17,18] as presented in Fig. 8. A constant value $\Delta a = a_{bcc-\mathrm{Gd}} - a_{bcc-\mathrm{Ti}} = 0.0775$ nm is derived from the room temperature values $a_{bcc\text{-Ti}} = 0.3275$ nm [18] and $a_{bcc\text{-Gd}} = 0.405$ nm [19] of pure bcc phases by assuming that the influence of slightly different thermal expansion coefficients can be neglected here. The lattice parameters of *bcc*-Ti(Gd) phase are only slightly increased, which is in accordance with a rather low solubility of Gd atoms. From the lattice parameter it follows that the Gd-content of *bcc*-Ti(Gd) is (0.6 ± 0.3) at % at $T = 1800$ K.

Analogously, the Ti content in the *bcc*-Gd(Ti) phase can be estimated from the lattice parameter $a_{bcc\text{-Gd(Ti)}}$:

$$x_{\mathrm{Ti}}(T) = \frac{a_{bcc-\mathrm{Gd(Ti)}}(T) - a_{bcc-\mathrm{Gd}}(T)}{a_{bcc-\mathrm{Ti}} - a_{bcc-\mathrm{Gd}}} 100. \quad (5)$$

Here, a value of $a_{bcc\text{-Gd}} = 0.4096$ nm at $T = 1500$ K is obtained by an extrapolation from the room temperature value of *bcc*-Gd ($a = 0.405$ nm [19]) along with the thermal expansion coefficient of Gd ($\alpha = 9.4\times10^{-6}$/K). The smaller value of the lattice parameter $a = (0.4092 \pm 0.0002)$ nm of the *bcc*-Gd(Ti) phase (Table 1) is due to dissolved Ti. Maximum solubility of (0.8 ± 0.3) at% Ti in *bcc*-Gd

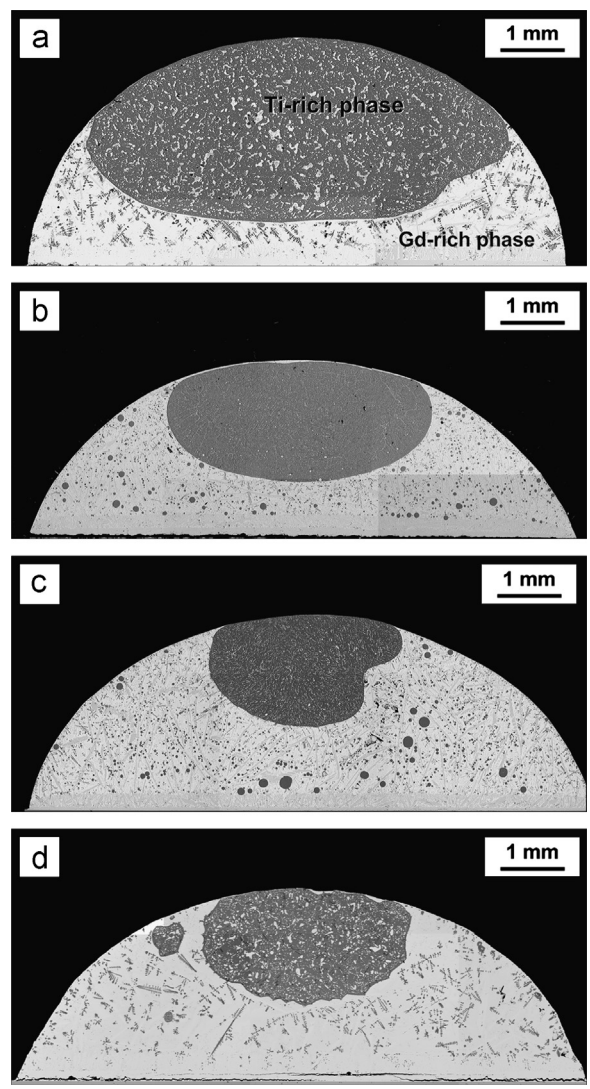

Fig. 9. SEM images of the cross section of solidified Gd–Ti alloys after sessel drop melting up to 2073 K (a) $Gd_{60}Ti_{40}$—slowly cooled, (b) $Gd_{40}Ti_{60}$—fast cooled, (c) $Gd_{70}Ti_{30}$—fast cooled, (d) $Gd_{70}Ti_{30}$—slowly cooled. Occurrence of liquid–liquid phase separation is concluded from the coarse microstructure. The Ti-enriched liquid (dark region) is lifted up due to the buoyancy resulting from the density differences. The (monotectic) chemical compositions of the liquids are within the error limits identical for the three samples (Table 1). The volume fractions of the two liquids change in accordance with the expectations from the lever rule. Secondary phase reaction and decomposition occur within the two former liquids upon quenching.

(Ti) at 1500 K are obtained from the diffraction data. Fig. 7 also comprises the results of phase composition and transformation temperatures as determined by DTA measurements, in situ XRD, and microstructure analysis discussed in the following section.

3.2. Microstructure of solidified Gd–Ti alloys

Fig. 9a–d shows the SEM images of complete cross sections of the different Gd–Ti samples after the sessile drop melt experiment. The optical observation of the melting behavior of the Gd–Ti alloys during heating up to $T=2073$ K was in good agreement with the in situ XRD measurements. In the sessile drop experiment the start of melting was at $T=(1495\pm20)$ K and the $Gd_{40}Ti_{60}$, $Gd_{60}Ti_{70}$ and $Gd_{70}Ti_{30}$ samples were completely molten at $T=(1870\pm50)$ K [20]. The comparison of the microstructures in Fig. 9a–d clearly reveals the occurrence of the decomposition into two liquids by a liquid phase separation $L\rightarrow L_1+L_2$. Due to the large difference in mass density, the lighter Ti-rich liquid L_1 (dark) is situated at the top of the sessile droplet. Corresponding to the lever rule the fraction of the Ti-rich liquid L_1 is decreased for higher Gd-content. The chemical compositions of the Ti-rich and Gd-rich regions were determined by EDX analysis over large areas of 500 μm × 500 μm. The measured values given in Table 2 are identical within the experimental errors and correspond to the compositions of the two liquids at the monotectic temperature. The microstructure of droplets cooled with different rates exhibit clear differences especially in their dendrites sizes; however there is no difference in the average chemical composition of the areas representing the former two liquids.

Upon cooling below the monotectic temperature the *bcc*-Ti(Gd) phase starts to solidify from the Ti-rich melt L_1. Fig. 10(a) shows the enlarged SEM image of the Ti-rich solidified liquid L_1 (dark region of Fig. 9d). According to the XRD results, the corresponding reaction scheme is $L_1\rightarrow$ *bcc*-Ti(Gd)+$L_2\rightarrow$ *bcc*-Ti (Gd)+*bcc*-Gd(Ti). The *bcc*-Ti(Gd) phase (dark) dissolves 0.5 at% Gd at the monotectic temperature as determined by the area analysis of corresponding regions of Fig. 10(a). With decreasing temperature Gd-phase becomes precipitated at the grain boundaries of the *bcc*-Ti (gray lines) due to the lowered solubility. The EDX point analysis delivers a solubility of 0.2 at% Gd in the *hcp*-Ti phase. The formation of the Gd-rich liquid L_2 by the monotectic reaction is evident by the round-shaped white areas in Fig. 10a. The contrast variation within the white areas is due to secondary Ti precipitation and the eutectic reaction within L_2 upon cooling.

Fig. 10(b) shows the enlarged SEM image the Gd-rich liquid (bright region of Fig. 9d) after solidification. The microstructure consists of Ti-dendrites (black) and regions with a typical Gd+Ti eutectic microstructure. According to the XRD results, the corresponding reaction scheme is $L_2\rightarrow$ *bcc*-Ti(Gd)+$L_2\rightarrow$ *bcc*-Ti(Gd)+ *bcc*-Gd(Ti). Gd-dendrites (gray) are also present formed according to the course of the liquidus line probably due to undercooling of the Gd-rich melt. For the eutectic composition a value of 92 ± 3 at% Gd is obtained from the EDX analysis of the corresponding Gd +Ti two-phase areas. About 0.2 at% Gd is found to be dissolved in the *hcp*-Ti(Gd) dendrites. On the other side about 0.5 at% Ti is dissolved in *hcp*-Gd(Ti) dendrites. The experimental values by SEM/EDX are summarized in Table 2 and also presented in Fig. 7.

Table 2
Chemical compositions of phases determined by SEM/EDX analysis.

	e_1		e_2		
	L_1: Gd (at%)	L_2: Gd (at%)	L: Gd (at%)	Gd: Ti (at%)	Ti:Gd (at%)
$Gd_{40}Ti_{60}$	7.0 ± 2	78 ± 5	91.4 ± 3	0.2 ± 0.1	0.4 ± 0.1
$Gd_{60}Ti_{40}$	5.5 ± 1	74 ± 5	–	0.2 ± 0.1	0.5 ± 0.1
$Gd_{70}Ti_{30}$	6.1 ± 1	75 ± 5	92.3 ± 2	0.3 ± 0.1	0.4 ± 0.1

4. Thermodynamic calculations

The CALPHAD method [21] was applied to assess thermodynamic parameters in the binary Gd–Ti system. The Gibbs energy of elements was defined with respect to its stable magnetically disordered state at 298.15 K and 10^5 Pa (stable element reference). The Gibbs energy of the elements was represented as a power series in terms of the temperature in the form:

$$^0G(T)=A+BT+CT\ln(T)+DT^2+ET^3+\frac{F}{T}+\sum_n G_nT^n, \quad (5)$$

where A to F and G_n are coefficients and n represents a set of integers. The parameters A to G_n were accepted from SGTE-Pure database 4.4 [22]. As reference state for the pure elements the Gibbs energy of hcp-Gd and hcp-Ti were used.

The liquid and the disordered solid solutions, bcc and hcp, were described by a substitutional solution model with the Gibbs energy being defined as

$$G^{\varphi}=\sum_{i=1}^{2}x_iG_i^{ref}+G^{id}+G^{ex};\qquad G^{id}=RT\sum_{i=1}^{2}x_i\ln(x_i)$$

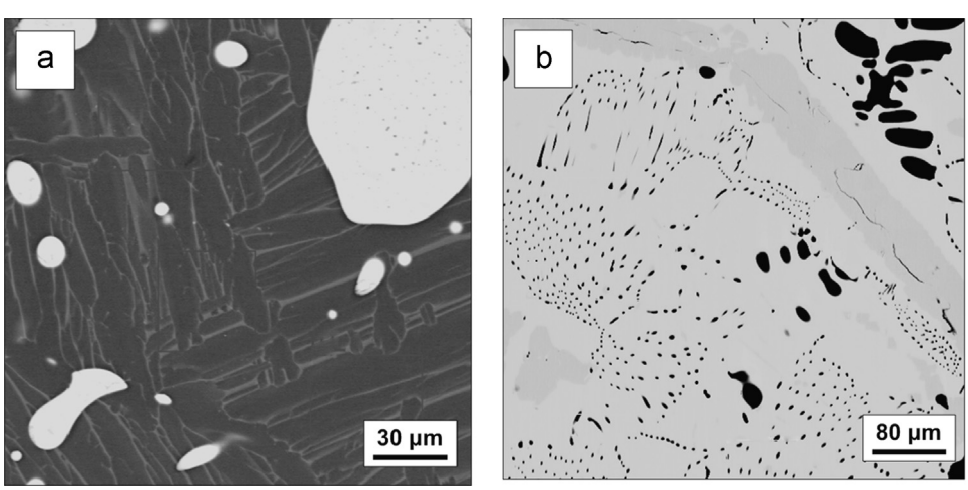

Fig. 10. Magnified SEM images of solidified Ti-rich (a) and Gd-rich (b) liquids of $Gd_{70}Ti_{30}$. (a) Ti-dendrites (black) with Gd precipitation at the grain boundary (gray lines) and solidified entrapped Gd-droplets (white) are visible. (b) Ti-dendrites (black) and (Gd+Ti) eutectic microstructure are visible.

Table 3
Summary of thermodynamic parameters describing the Gd–Ti system referred to the stable element reference H^{SER} (T=298.15 K, P=100 kPa). Values are given in SI units (J, mol, and K), except β (μ_B per mole of atoms).

Parameter	Ref.
G_{Gd}^{hcp}, G_{Gd}^{bcc}, G_{Gd}^{liquid}, G_{Ti}^{hcp}, G_{Ti}^{bcc}, G_{Ti}^{hcp}, G_{Ti}^{liquid}	[22]
Liquid	
${}^0L_{Gd,Ti}^{liq} = 93481 - 30.90272T$	This work
${}^1L_{Gd,Ti}^{liq} = -8955$	This work
hcp	
${}^0L_{Gd,Ti}^{hcp} = +44248.7 + 19.9839T$	This work
${}^1L_{Gd,Ti}^{hcp} = 2176.21$	This work
bcc	
${}^0L_{Gd,Ti}^{bcc} = +39476.4 + 15.687T$	This work
${}^1L_{Gd,Ti}^{bcc} = +2152.38$	This work
Magnetic parameter $Tc_{Gd} = 293.4$, $\beta_{Gd} = 3$	[22]

$$G^{ex} = \sum_{i=1}^{2} \sum_{j>i}^{2} x_i x_j \sum_{v=0}^{n} {}^{v}L_{i,j}(x_i - x_j)^v \quad (6)$$

where G_i^{ref} was the reference Gibbs energy of the pure element i, i.e., Gd or Ti in the corresponding phase φ, G^{id} was the ideal mixing contribution, G^{ex} was the excess energy expressed by the Redlich–Kister polynomial function, x_i and x_j are the mole fractions of the constituents i, and j, respectively. The parameters ${}^{v}L_{i,j}$ were the binary interaction parameters between components i and j of an order ν.

Magnetic contributions to the Gibbs energy of the *hcp*-Gd phase were taken into account according to the treatment proposed by Hillert and Jarl [23]. For a phase with the magnetic moment β (Bohr's magneton per mole of atoms) and the magnetic ordering at Curie or Neel temperature T_C, the magnetic contribution to its Gibbs energy is expressed as [24]

$$G^{mag} = RT \ln(\beta + 1) f(T/T_C) \quad (7)$$

where $f(T/T_C)$ represents a polynomial function. For bcc-Ti, bcc-Gd, and hcp-Ti, the magnetic moment β was fixed as value equal to zero.

The thermodynamic model parameters for the binary system Gd–Ti were assessed using the least-squares optimization module PARROT of Thermo-Calc software [25]. The experimentally determined temperatures and phase compositions for invariant reactions given in Table 1 were used for the optimization at the first stage. Then data for liquidus, *bcc*-Gd solvus and *hcp*-Gd solvus were used at the second step of optimization. It should be mentioned that the experimental data of Tedenac et al. [26] for mixing enthalpy of liquid measured by isoperibolic calorimetry were not used in optimization because the determined enthalpies of mixing were negative values that could not be consistent with phase diagram having miscibility gap in liquid phase. Additionally it should be noted that coordinates of maximal point of miscibility gap (temperature and composition) play important role in constraining mixing parameters of liquid and determination of this point could be recommended for future experimental investigations. The final thermodynamic parameters derived in the present work are given in Table 3. These thermodynamic parameters were used to compute the Gd–Ti phase diagram shown in Fig. 7 along with the experimental results from the present study.

5. Discussion

According to calculations, the reaction *bcc*-Ti(Gd)↔*hcp*-Gd(Ti) +*hcp*-Ti(Gd) has eutectoid character. It should be mentioned that the authors of Ref. [2] concluded that the character of this reaction is peritectoid based on composition of *hcp*-Ti (4 at% Gd), while the temperature of this reaction was within some scatter below the transformation *bcc*↔*hcp* in pure Ti thus indicating eutectoid character of this reaction. It was mentioned [2] that the uncertainty of the temperature measurements by DTA and resistivity made it impossible to decide about the character of this reaction and the nature of this reaction. However much less solubilities were reported in Ref. [3]. Also the corresponding SEM image in Ref. [2] is not that of a single *hcp*-Ti phase. The composition of *bcc* and *hcp* phases determined in the present work was close to each other and pure Ti. Therefore it was impossible to determine the character of this reaction based on the compositions. However the temperature of *bcc*-Ti(Gd)↔*hcp*-Gd(Ti)+*hcp*-Ti(Gd) reaction determined by DTA in the present work is below transition of *bcc*↔*hcp* in pure Ti accepted for calculations according to Ref. [22] thus indicating the eutectoid character of the above mentioned reaction. The value of the eutectic composition of the reaction *L*↔*bcc*-Gd(Ti)+*bcc*-Ti(Gd) determined by area SEM/EDX analysis for different samples and sample regions exhibit some scatter ranging from 89 to 95 at%. The average value is given in Table 1. The calculation of the liquidus line by Eq. (3) can be used as a first approximation. Because no experimental data for the Gd-rich liquid are available to prove we cannot exclude some additional systematic deviations which could be the reason for the discrepancies with the calculated phase diagram. Taking into account the experimental error especially for the liquidus line, the result of the present modeling agrees well with the experimental data. The calculated temperature and phase compositions for invariant equilibria are given in comparison with experimental data in Table 1.

6. Conclusions

An experimental and thermodynamic assessment of the binary Gd–Ti system has been carried out. By combining electrostatic levitation with high-energy X-ray diffraction the phase equilibria could be studied in situ at elevated temperatures. The coexistence ranges of the phases at high temperature could directly be determined. In particular, an extended range of coexistence of *bcc*-Ti and Gd-rich liquid phase up to 1840 K was detected not known from previous assessments [1–5]. The experimental observations led to an improved new Gd–Ti phase diagram. Based on the experimental results, a consistent thermodynamic description was developed. Our results show, that the assessment of phase diagrams can get great advantages by using in situ X-ray diffraction and container-less sample processing especially for alloy systems with high liquidus temperature, and/or reactive melts.

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Appendix A. Supplementary material

Supplementary materials associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.06.007.

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